

# V3 empirical study · Instruction

Seven models · four underlyings · four volatility regimes · 1.5-year fixed calibration

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## What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 7 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

## Experimental design (held fixed for every model)

- Models: GBM, Modified GBM (Markov direction), GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH-Merton, Heston-Merton (Bates).
- Modified GBM (Markov direction + split-normal magnitudes) is specified in `V3_modified_gbm_model.pdf` in this folder.
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at  $t_0$  are used only after  $t_0$ .
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

# V3 empirical study · Instruction (continued)

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## Shared market sample and filters (not model-specific)

- For each company  $\times$  regime, one listed-call sample is drawn once and reused by all seven models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite  $S$ ,  $K$ ,  $C$ ; (2) no-arbitrage  $C \geq \max(0, S-K)$ ; (3)  $7 \leq \text{DTE} \leq 60$ ; (4)  $|S/K - 1| \leq 10\%$ ; (5) premium  $\geq 0.05$ , valid bid-ask with relative spread  $\leq 50\%$  when those columns exist, volume  $\geq 1$  when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

## Model-correctness adjustments (V3)

- Heston / Heston–Merton:  $(\kappa, \theta, \xi, \rho, v_0)$  are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments.  $\mu$  is the P-measure lookback mean of stock returns; LSM replaces  $\mu \rightarrow r$ .
- Euler ordering: the stock increment over  $[t, t+\Delta t]$  uses the current variance  $v_t$ ;  $v_{t+\Delta t}$  is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ( $n_{\text{paths}} = 10000$ , seed 42,  $n_{\text{steps}} = \text{DTE}$ ,  $\Delta t = 1/252$ ). Paths are not averaged before stopping.

## Metrics

- Primary:  $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$ . Ranking uses this number only.
- Secondary:  $\text{MAE} = \text{mean} |C_{\text{model}} - C_{\text{mkt}}|$  (dollars).  $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$  (dollars).  $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$ . Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

## Page 3 · Filters, sample, and no-look-ahead

### Shared evaluation sample · n listed calls (identical for all seven models in this report)

| Regime               | Window                  | SPY n | AAPL n | MSFT n | AMZN n |
|----------------------|-------------------------|-------|--------|--------|--------|
| 2008-2009 Crisis     | 2008-08-01 → 2009-07-31 | 50    | 50     | 50     | 50     |
| 2013-2014 Normal     | 2014-01-01 → 2014-12-31 | 50    | 50     | 50     | 50     |
| 2018-2019 Late-cycle | 2018-10-01 → 2019-09-30 | 50    | 51     | 50     | 51     |
| 2019-2020 COVID      | 2019-09-01 → 2020-08-31 | 50    | 50     | 50     | 50     |

### Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

| Ticker | Regime    | Input* | No-arb | DTE 7-60 | S/K-1 ≤10% | Liquidity |
|--------|-----------|--------|--------|----------|------------|-----------|
| SPY    | 2008-2009 | 3359   | 3311   | 3311     | 3203       | 3203      |
| SPY    | 2013-2014 | 12815  | 12548  | 12548    | 12405      | 12405     |
| SPY    | 2018-2019 | 35222  | 35104  | 35104    | 34809      | 34809     |
| SPY    | 2019-2020 | 42100  | 41961  | 41961    | 41379      | 41379     |
| AAPL   | 2008-2009 | 685    | 685    | 685      | 652        | 652       |
| AAPL   | 2013-2014 | 8925   | 8875   | 8875     | 8633       | 8633      |
| AAPL   | 2018-2019 | 6309   | 6297   | 6297     | 6084       | 6084      |
| AAPL   | 2019-2020 | 8010   | 7992   | 7992     | 7713       | 7713      |
| MSFT   | 2008-2009 | 566    | 566    | 566      | 538        | 538       |
| MSFT   | 2013-2014 | 3299   | 3237   | 3237     | 3175       | 3175      |
| MSFT   | 2018-2019 | 7814   | 7752   | 7752     | 7583       | 7583      |
| MSFT   | 2019-2020 | 6892   | 6852   | 6852     | 6655       | 6655      |
| AMZN   | 2008-2009 | 465    | 465    | 465      | 442        | 442       |
| AMZN   | 2013-2014 | 5715   | 5692   | 5692     | 5598       | 5598      |
| AMZN   | 2018-2019 | 28809  | 28808  | 28808    | 28346      | 28346     |
| AMZN   | 2019-2020 | 26766  | 26760  | 26760    | 26323      | 26323     |

\*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date  $t$  use only equity returns and listed quotes with  $\text{trading\_date} \leq t$ . LSM prices the Monday contract using the latest calibration row with date  $\leq$  that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so  $n$  matches AAPL/MSFT (~50). All seven models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to  $\leq 24$  contracts (maturity  $\times$  moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps:  $3\sigma$  residual threshold where applicable).

## Page 4 · Table block · SPY · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n\_paths=10000, seed=42.

**Table SPY-1 SPY · 2008-2009 Crisis**  
2008-08-01 → 2009-07-31 · n = 50 contracts

| Model               | RMSE%        | MAE           | Bias \$        | Bias%         | Early-ex. frac. | Mark     |
|---------------------|--------------|---------------|----------------|---------------|-----------------|----------|
| GBM                 | 39.85        | 1.5118        | -1.3570        | -29.27        | 25.5%           | 5        |
| Modified GBM        | 43.13        | 1.6255        | -1.5635        | -35.30        | 26.1%           | 7        |
| GARCH               | 36.15        | 1.3736        | -0.9849        | -18.26        | 23.5%           | 3        |
| Heston              | 39.13        | 1.4774        | -1.2428        | -26.22        | 18.1%           | 4        |
| Merton              | 40.12        | 1.5229        | -1.3790        | -29.99        | 24.9%           | 6        |
| <b>GARCH-Merton</b> | <b>35.69</b> | <b>1.3398</b> | <b>-0.8186</b> | <b>-13.54</b> | <b>20.8%</b>    | <b>1</b> |
| Heston-Merton       | 35.79        | 1.3326        | -0.8124        | -13.60        | 7.9%            | 2        |

**Table SPY-2 SPY · 2013-2014 Normal**  
2014-01-01 → 2014-12-31 · n = 50 contracts

| Model         | RMSE%        | MAE           | Bias \$        | Bias%         | Early-ex. frac. | Mark     |
|---------------|--------------|---------------|----------------|---------------|-----------------|----------|
| GBM           | 40.10        | 0.7145        | +0.5358        | +27.51        | 26.2%           | 2        |
| Modified GBM  | 41.67        | 0.7464        | +0.5698        | +29.03        | 26.1%           | 4        |
| <b>GARCH</b>  | <b>38.23</b> | <b>0.6786</b> | <b>+0.4831</b> | <b>+25.30</b> | <b>24.6%</b>    | <b>1</b> |
| Heston        | 45.98        | 0.8231        | +0.6985        | +34.22        | 19.2%           | 6        |
| Merton        | 40.53        | 0.7231        | +0.5464        | +27.95        | 24.2%           | 3        |
| GARCH-Merton  | 45.88        | 0.8274        | +0.6986        | +34.18        | 22.4%           | 5        |
| Heston-Merton | 77.92        | 1.5212        | +1.4925        | +67.23        | 13.4%           | 7        |

**Table SPY-3 SPY · 2018-2019 Late-cycle**  
2018-10-01 → 2019-09-30 · n = 50 contracts

| Model         | RMSE%        | MAE           | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|---------------|--------------|---------------|----------------|--------------|-----------------|----------|
| GBM           | 23.62        | 0.8840        | -0.1277        | +1.81        | 20.8%           | 4        |
| Modified GBM  | 23.03        | 0.8902        | -0.2283        | -0.59        | 22.2%           | 3        |
| GARCH         | 25.60        | 1.0389        | -0.8655        | -16.06       | 18.9%           | 5        |
| Heston        | 27.91        | 0.9468        | +0.2281        | +10.44       | 16.2%           | 6        |
| <b>Merton</b> | <b>22.81</b> | <b>0.9091</b> | <b>-0.4865</b> | <b>-6.72</b> | <b>17.7%</b>    | <b>1</b> |
| GARCH-Merton  | 22.86        | 0.9032        | -0.4350        | -5.93        | 16.7%           | 2        |
| Heston-Merton | 38.79        | 1.2360        | +0.8907        | +25.18       | 14.2%           | 7        |

**Table SPY-4 SPY · 2019-2020 COVID**  
2019-09-01 → 2020-08-31 · n = 50 contracts

| Model         | RMSE%        | MAE           | Bias \$        | Bias%         | Early-ex. frac. | Mark     |
|---------------|--------------|---------------|----------------|---------------|-----------------|----------|
| GBM           | 57.52        | 2.9208        | -0.5111        | +18.64        | 24.3%           | 3        |
| Modified GBM  | 59.99        | 2.9847        | -0.3345        | +21.88        | 23.9%           | 4        |
| GARCH         | 60.68        | 2.9733        | -0.3077        | +22.51        | 22.9%           | 5        |
| <b>Heston</b> | <b>53.99</b> | <b>2.9171</b> | <b>-0.9629</b> | <b>+11.31</b> | <b>19.5%</b>    | <b>1</b> |
| Merton        | 56.81        | 2.9399        | -0.5830        | +17.28        | 20.3%           | 2        |
| GARCH-Merton  | 70.25        | 3.1595        | +0.2869        | +33.51        | 20.3%           | 7        |
| Heston-Merton | 69.93        | 3.2104        | +0.3530        | +34.09        | 11.4%           | 6        |

RMSE% is  $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$ . MAE and Bias \$ are in option-premium dollars. Bias% is  $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$ . Early-ex. frac. is the mean LSM share of paths e

## Page 5 · Table block · AAPL · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n\_paths=10000, seed=42.

**Table AAPL-1 AAPL · 2008-2009 Crisis**  
2008-08-01 → 2009-07-31 · n = 50 contracts

| Model               | RMSE%        | MAE           | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|---------------------|--------------|---------------|----------------|--------------|-----------------|----------|
| GBM                 | 34.70        | 1.7325        | -0.2012        | +2.49        | 23.4%           | 4        |
| <b>Modified GBM</b> | <b>32.72</b> | <b>1.6706</b> | <b>-0.6554</b> | <b>-4.84</b> | <b>24.4%</b>    | <b>1</b> |
| GARCH               | 35.04        | 1.7531        | -0.1784        | +2.83        | 21.8%           | 6        |
| Heston              | 33.29        | 1.7314        | -0.4465        | -1.87        | 18.9%           | 2        |
| Merton              | 34.57        | 1.7293        | -0.2406        | +1.71        | 21.0%           | 3        |
| GARCH-Merton        | 43.53        | 2.0289        | +0.7015        | +16.92       | 16.7%           | 7        |
| Heston-Merton       | 34.75        | 1.7486        | -0.1335        | +3.29        | 12.0%           | 5        |

**Table AAPL-2 AAPL · 2013-2014 Normal**  
2014-01-01 → 2014-12-31 · n = 50 contracts

| Model         | RMSE%        | MAE           | Bias \$        | Bias%         | Early-ex. frac. | Mark     |
|---------------|--------------|---------------|----------------|---------------|-----------------|----------|
| GBM           | 57.70        | 3.8194        | +3.8194        | +51.38        | 24.4%           | 4        |
| Modified GBM  | 51.10        | 3.3484        | +3.3484        | +44.85        | 25.7%           | 2        |
| <b>GARCH</b>  | <b>39.62</b> | <b>2.4884</b> | <b>+2.3946</b> | <b>+32.33</b> | <b>23.1%</b>    | <b>1</b> |
| Heston        | 70.88        | 4.8769        | +4.8769        | +64.15        | 19.2%           | 6        |
| Merton        | 57.50        | 3.8718        | +3.8718        | +51.52        | 19.1%           | 3        |
| GARCH-Merton  | 67.13        | 4.6054        | +4.6054        | +61.25        | 17.2%           | 5        |
| Heston-Merton | 73.89        | 5.0844        | +5.0844        | +67.12        | 16.9%           | 7        |

**Table AAPL-3 AAPL · 2018-2019 Late-cycle**  
2018-10-01 → 2019-09-30 · n = 51 contracts

| Model         | RMSE%        | MAE           | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|---------------|--------------|---------------|----------------|--------------|-----------------|----------|
| <b>GBM</b>    | <b>16.77</b> | <b>0.8809</b> | <b>-0.5584</b> | <b>-7.11</b> | <b>23.8%</b>    | <b>1</b> |
| Modified GBM  | 16.88        | 0.8919        | -0.5478        | -6.91        | 24.5%           | 2        |
| GARCH         | 17.11        | 0.9076        | -0.6227        | -8.15        | 22.6%           | 3        |
| Heston        | 19.55        | 0.9559        | +0.3612        | +7.54        | 14.6%           | 6        |
| Merton        | 18.08        | 0.8489        | +0.2167        | +5.33        | 14.1%           | 5        |
| GARCH-Merton  | 17.34        | 0.8370        | +0.0909        | +3.31        | 19.3%           | 4        |
| Heston-Merton | 29.95        | 1.4985        | +1.2784        | +22.29       | 14.0%           | 7        |

**Table AAPL-4 AAPL · 2019-2020 COVID**  
2019-09-01 → 2020-08-31 · n = 50 contracts

| Model         | RMSE%        | MAE           | Bias \$        | Bias%         | Early-ex. frac. | Mark     |
|---------------|--------------|---------------|----------------|---------------|-----------------|----------|
| <b>GBM</b>    | <b>41.39</b> | <b>3.1111</b> | <b>+0.6446</b> | <b>+18.83</b> | <b>24.0%</b>    | <b>1</b> |
| Modified GBM  | 43.50        | 3.2399        | +0.9173        | +21.54        | 24.2%           | 2        |
| GARCH         | 45.91        | 3.3764        | +1.2410        | +24.56        | 22.0%           | 5        |
| Heston        | 44.31        | 3.2876        | +1.0658        | +22.80        | 19.5%           | 4        |
| Merton        | 43.70        | 3.2565        | +0.9786        | +22.02        | 17.8%           | 3        |
| GARCH-Merton  | 68.46        | 5.0306        | +3.8961        | +50.08        | 18.7%           | 7        |
| Heston-Merton | 51.81        | 3.7424        | +1.9950        | +31.82        | 10.0%           | 6        |

## Page 6 · Table block · MSFT · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n\_paths=10000, seed=42.

**Table MSFT-1 MSFT · 2008-2009 Crisis**  
2008-08-01 → 2009-07-31 · n = 50 contracts

| Model                | RMSE%        | MAE           | Bias \$        | Bias%         | Early-ex. frac. | Mark     |
|----------------------|--------------|---------------|----------------|---------------|-----------------|----------|
| GBM                  | 29.17        | 0.3120        | -0.2729        | -19.88        | 25.5%           | 6        |
| Modified GBM         | 28.96        | 0.3119        | -0.2681        | -19.42        | 26.3%           | 5        |
| GARCH                | 26.24        | 0.2848        | -0.0629        | +0.36         | 23.1%           | 3        |
| Heston               | 26.56        | 0.2969        | -0.2098        | -13.71        | 20.8%           | 4        |
| Merton               | 26.16        | 0.2932        | -0.1837        | -11.78        | 18.4%           | 2        |
| GARCH-Merton         | 32.88        | 0.2963        | +0.0849        | +14.04        | 20.0%           | 7        |
| <b>Heston-Merton</b> | <b>25.66</b> | <b>0.2922</b> | <b>-0.1697</b> | <b>-10.06</b> | <b>14.9%</b>    | <b>1</b> |

**Table MSFT-2 MSFT · 2013-2014 Normal**  
2014-01-01 → 2014-12-31 · n = 50 contracts

| Model         | RMSE%        | MAE           | Bias \$        | Bias%         | Early-ex. frac. | Mark     |
|---------------|--------------|---------------|----------------|---------------|-----------------|----------|
| GBM           | 62.49        | 0.4284        | +0.4262        | +52.88        | 24.7%           | 3        |
| Modified GBM  | 60.58        | 0.4138        | +0.4100        | +51.07        | 26.0%           | 2        |
| GARCH         | 64.23        | 0.4441        | +0.4431        | +54.76        | 23.9%           | 4        |
| Heston        | 76.15        | 0.5487        | +0.5487        | +66.83        | 18.7%           | 5        |
| <b>Merton</b> | <b>55.50</b> | <b>0.3746</b> | <b>+0.3666</b> | <b>+45.96</b> | <b>17.5%</b>    | <b>1</b> |
| GARCH-Merton  | 96.85        | 0.7307        | +0.7307        | +87.77        | 17.1%           | 7        |
| Heston-Merton | 80.64        | 0.5846        | +0.5846        | +71.12        | 16.5%           | 6        |

**Table MSFT-3 MSFT · 2018-2019 Late-cycle**  
2018-10-01 → 2019-09-30 · n = 50 contracts

| Model               | RMSE%        | MAE           | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|---------------------|--------------|---------------|----------------|--------------|-----------------|----------|
| GBM                 | 21.68        | 0.5626        | +0.0129        | +3.25        | 23.0%           | 3        |
| <b>Modified GBM</b> | <b>19.94</b> | <b>0.5524</b> | <b>-0.2155</b> | <b>-3.65</b> | <b>25.4%</b>    | <b>1</b> |
| GARCH               | 20.42        | 0.5496        | -0.1917        | -2.97        | 21.3%           | 2        |
| Heston              | 23.91        | 0.5954        | +0.1129        | +6.13        | 14.7%           | 4        |
| Merton              | 25.35        | 0.6315        | +0.2531        | +10.52       | 12.3%           | 5        |
| GARCH-Merton        | 25.36        | 0.6383        | +0.2795        | +11.22       | 16.3%           | 6        |
| Heston-Merton       | 42.61        | 1.0877        | +0.9716        | +32.04       | 10.1%           | 7        |

**Table MSFT-4 MSFT · 2019-2020 COVID**  
2019-09-01 → 2020-08-31 · n = 50 contracts

| Model               | RMSE%        | MAE           | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|---------------------|--------------|---------------|----------------|--------------|-----------------|----------|
| GBM                 | 45.98        | 1.7242        | +0.0264        | +16.93       | 24.8%           | 4        |
| <b>Modified GBM</b> | <b>41.04</b> | <b>1.7043</b> | <b>-0.3996</b> | <b>+8.84</b> | <b>25.5%</b>    | <b>1</b> |
| GARCH               | 42.17        | 1.6816        | -0.2778        | +11.08       | 22.6%           | 2        |
| Heston              | 42.53        | 1.7172        | -0.2236        | +12.09       | 18.0%           | 3        |
| Merton              | 46.60        | 1.7641        | +0.0420        | +17.39       | 21.5%           | 5        |
| GARCH-Merton        | 53.39        | 1.8833        | +0.4719        | +25.84       | 18.4%           | 6        |
| Heston-Merton       | 55.19        | 1.9181        | +0.6024        | +28.24       | 7.4%            | 7        |

## Page 7 · Table block · AMZN · seven models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n\_paths=10000, seed=42.

**Table AMZN-1 AMZN · 2008-2009 Crisis**  
2008-08-01 → 2009-07-31 · n = 50 contracts

| Model                | RMSE%        | MAE           | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|----------------------|--------------|---------------|----------------|--------------|-----------------|----------|
| GBM                  | 32.38        | 1.2336        | -0.0334        | +3.64        | 22.9%           | 6        |
| Modified GBM         | 29.85        | 1.2017        | -0.4075        | -4.77        | 25.2%           | 4        |
| GARCH                | 32.34        | 1.2318        | -0.0435        | +3.42        | 22.3%           | 5        |
| Heston               | 29.76        | 1.2209        | -0.6136        | -9.76        | 17.8%           | 3        |
| Merton               | 29.53        | 1.2007        | -0.3350        | -3.61        | 16.7%           | 2        |
| GARCH-Merton         | 53.16        | 1.7374        | +1.2478        | +32.42       | 15.6%           | 7        |
| <b>Heston-Merton</b> | <b>29.49</b> | <b>1.2180</b> | <b>-0.3014</b> | <b>-3.12</b> | <b>9.2%</b>     | <b>1</b> |

**Table AMZN-2 AMZN · 2013-2014 Normal**  
2014-01-01 → 2014-12-31 · n = 50 contracts

| Model         | RMSE%        | MAE           | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|---------------|--------------|---------------|----------------|--------------|-----------------|----------|
| GBM           | 23.09        | 2.4183        | +0.0489        | +5.54        | 24.8%           | 4        |
| Modified GBM  | 22.52        | 2.3818        | -0.0168        | +4.90        | 25.5%           | 2        |
| GARCH         | 22.58        | 2.3967        | -0.0514        | +4.59        | 22.8%           | 3        |
| Heston        | 41.35        | 3.4101        | +2.9636        | +30.50       | 16.0%           | 6        |
| <b>Merton</b> | <b>22.30</b> | <b>2.3829</b> | <b>-0.1145</b> | <b>+4.05</b> | <b>19.7%</b>    | <b>1</b> |
| GARCH-Merton  | 35.61        | 2.9530        | +2.1822        | +24.10       | 18.2%           | 5        |
| Heston-Merton | 51.11        | 4.3555        | +4.1435        | +41.10       | 13.6%           | 7        |

**Table AMZN-3 AMZN · 2018-2019 Late-cycle**  
2018-10-01 → 2019-09-30 · n = 51 contracts

| Model         | RMSE%        | MAE            | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|---------------|--------------|----------------|----------------|--------------|-----------------|----------|
| <b>GBM</b>    | <b>22.97</b> | <b>12.4353</b> | <b>-5.5218</b> | <b>-4.77</b> | <b>22.9%</b>    | <b>1</b> |
| Modified GBM  | 23.72        | 12.4787        | -3.3465        | -1.02        | 21.5%           | 4        |
| GARCH         | 23.04        | 12.4406        | -5.4598        | -4.84        | 21.5%           | 2        |
| Heston        | 32.20        | 15.3154        | +6.3787        | +13.43       | 14.0%           | 6        |
| Merton        | 23.53        | 12.7074        | -7.4064        | -8.19        | 14.9%           | 3        |
| GARCH-Merton  | 28.66        | 13.9181        | +3.7835        | +10.56       | 14.3%           | 5        |
| Heston-Merton | 57.83        | 26.7876        | +24.5087       | +43.45       | 9.3%            | 7        |

**Table AMZN-4 AMZN · 2019-2020 COVID**  
2019-09-01 → 2020-08-31 · n = 50 contracts

| Model         | RMSE%        | MAE            | Bias \$        | Bias%        | Early-ex. frac. | Mark     |
|---------------|--------------|----------------|----------------|--------------|-----------------|----------|
| GBM           | 56.86        | 26.5091        | +13.6258       | +35.54       | 23.7%           | 5        |
| Modified GBM  | 60.74        | 27.6000        | +16.4384       | +39.62       | 22.9%           | 7        |
| GARCH         | 37.53        | 22.7400        | -4.0628        | +10.43       | 22.5%           | 2        |
| <b>Heston</b> | <b>37.33</b> | <b>22.8307</b> | <b>-4.3221</b> | <b>+9.37</b> | <b>16.5%</b>    | <b>1</b> |
| Merton        | 58.57        | 27.0617        | +15.3098       | +37.68       | 20.1%           | 6        |
| GARCH-Merton  | 44.57        | 23.4896        | +3.5116        | +20.71       | 21.1%           | 3        |
| Heston-Merton | 46.06        | 23.8261        | +4.7029        | +22.38       | 8.8%            | 4        |

## Page 7 · Summary tables · RMSE% across companies and regimes

**Table S1 · Overall ranking (mean RMSE% over 16 cells)**

| Rank | Model         | Mean RMSE% | Median RMSE% | # of 16 cells best |
|------|---------------|------------|--------------|--------------------|
| 1    | GARCH         | 35.43      | 35.59        | 2                  |
| 2    | Modified GBM  | 37.46      | 36.88        | 3                  |
| 3    | Merton        | 37.60      | 37.34        | 3                  |
| 4    | GBM           | 37.89      | 37.28        | 3                  |
| 5    | Heston        | 40.30      | 38.23        | 2                  |
| 6    | GARCH-Merton  | 46.35      | 44.05        | 1                  |
| 7    | Heston-Merton | 50.09      | 48.58        | 2                  |

**Table S2 · Wins by company (lowest RMSE% in that company×regime)**

| Model         | SPY wins | AAPL wins | MSFT wins | AMZN wins | Total |
|---------------|----------|-----------|-----------|-----------|-------|
| GBM           | 0        | 2         | 0         | 1         | 3     |
| Modified GBM  | 0        | 1         | 2         | 0         | 3     |
| GARCH         | 1        | 1         | 0         | 0         | 2     |
| Heston        | 1        | 0         | 0         | 1         | 2     |
| Merton        | 1        | 0         | 1         | 1         | 3     |
| GARCH-Merton  | 1        | 0         | 0         | 0         | 1     |
| Heston-Merton | 0        | 0         | 1         | 1         | 2     |

**Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)**

| Model         | 2008-2009<br>Crisis | 2013-2014<br>Normal | 2018-2019<br>Late-cycle | 2019-2020<br>COVID | Mean of 4 |
|---------------|---------------------|---------------------|-------------------------|--------------------|-----------|
| GBM           | 34.03               | 45.85               | 21.26                   | 50.44              | 37.89     |
| Modified GBM  | 33.66               | 43.97               | 20.89                   | 51.32              | 37.46     |
| GARCH         | 32.44               | 41.17               | 21.54                   | 46.57              | 35.43     |
| Heston        | 32.19               | 58.59               | 25.89                   | 44.54              | 40.30     |
| Merton        | 32.59               | 43.96               | 22.44                   | 51.42              | 37.60     |
| GARCH-Merton  | 41.32               | 61.37               | 23.55                   | 59.17              | 46.35     |
| Heston-Merton | 31.42               | 70.89               | 42.30                   | 55.75              | 50.09     |

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).



Table S4.1 · SPY

| Model         | Mean RMSE%   | Median       | # regimes best |
|---------------|--------------|--------------|----------------|
| GBM           | 40.27        | 39.98        | 0              |
| Modified GBM  | 41.95        | 42.40        | 0              |
| GARCH         | 40.17        | 37.19        | 1              |
| Heston        | 41.75        | 42.55        | 1              |
| <b>Merton</b> | <b>40.07</b> | <b>40.32</b> | <b>1</b>       |
| GARCH-Merton  | 43.67        | 40.78        | 1              |
| Heston-Merton | 55.61        | 54.36        | 0              |

Table S4.2 · AAPL

| Model         | Mean RMSE%   | Median       | # regimes best |
|---------------|--------------|--------------|----------------|
| GBM           | 37.64        | 38.04        | 2              |
| Modified GBM  | 36.05        | 38.11        | 1              |
| <b>GARCH</b>  | <b>34.42</b> | <b>37.33</b> | <b>1</b>       |
| Heston        | 42.01        | 38.80        | 0              |
| Merton        | 38.46        | 39.13        | 0              |
| GARCH-Merton  | 49.12        | 55.33        | 0              |
| Heston-Merton | 47.60        | 43.28        | 0              |

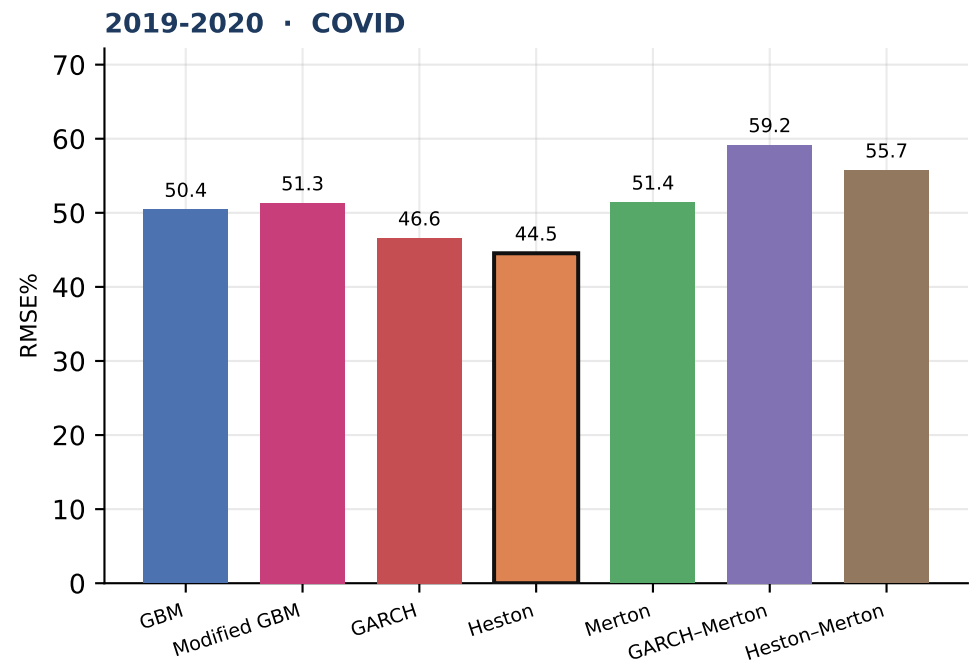
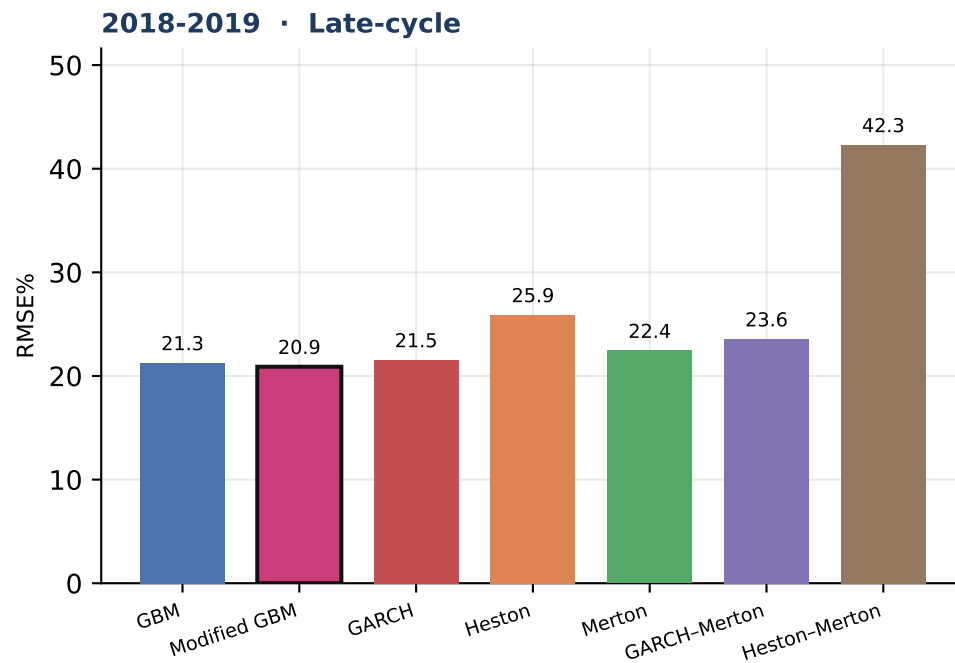
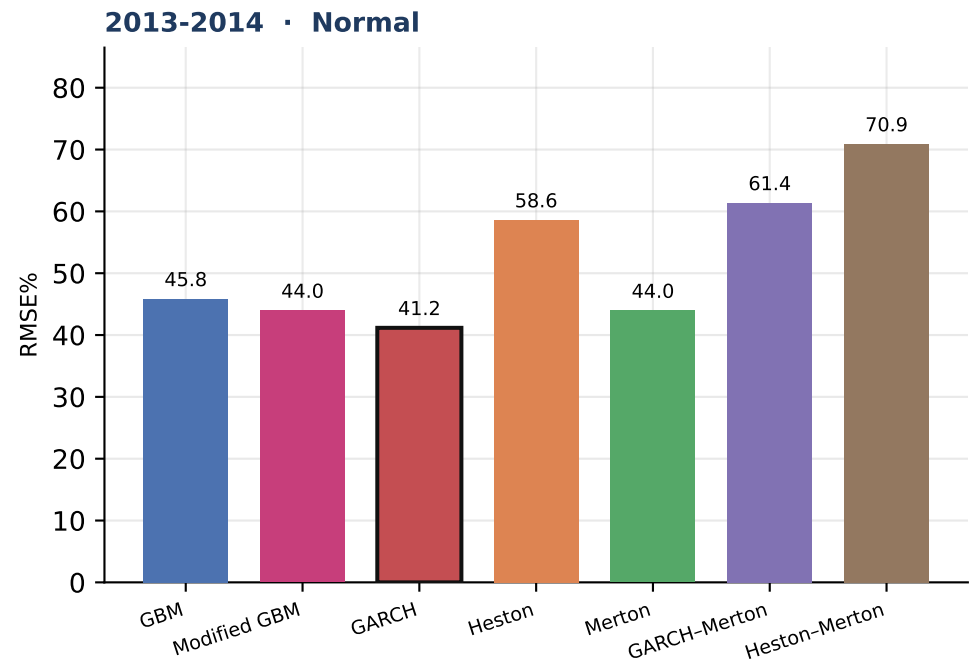
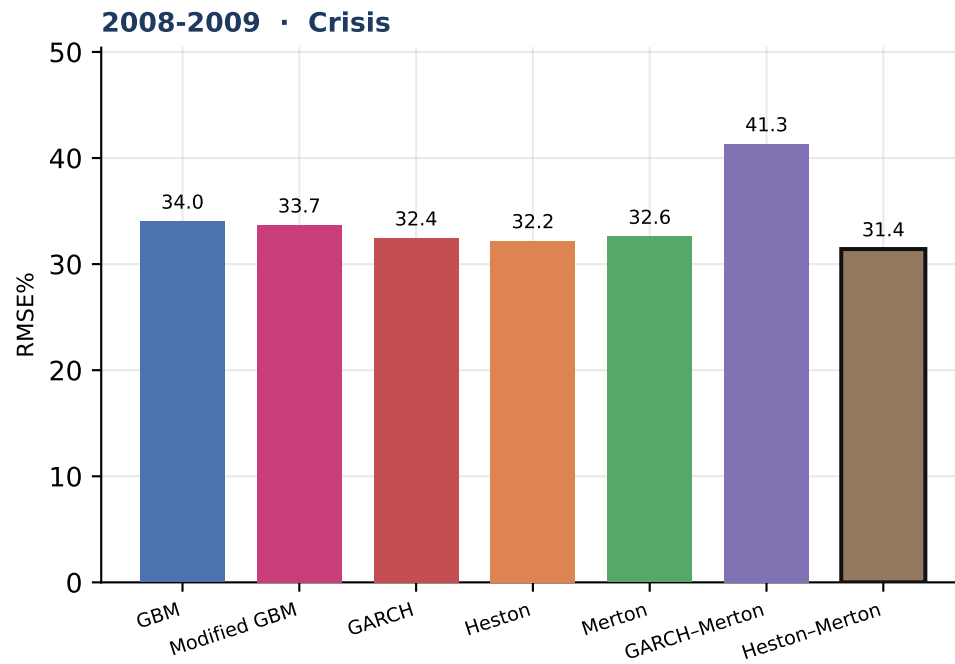
Table S4.3 · MSFT

| Model               | Mean RMSE%   | Median       | # regimes best |
|---------------------|--------------|--------------|----------------|
| GBM                 | 39.83        | 37.57        | 0              |
| <b>Modified GBM</b> | <b>37.63</b> | <b>35.00</b> | <b>2</b>       |
| GARCH               | 38.27        | 34.21        | 0              |
| Heston              | 42.29        | 34.55        | 0              |
| Merton              | 38.40        | 36.38        | 1              |
| GARCH-Merton        | 52.12        | 43.14        | 0              |
| Heston-Merton       | 51.02        | 48.90        | 1              |

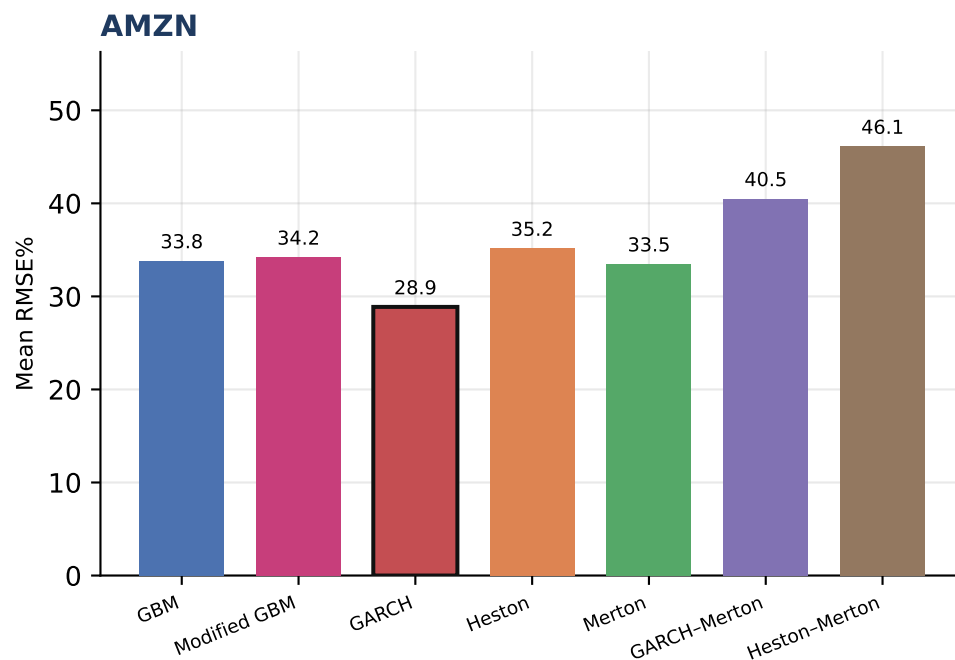
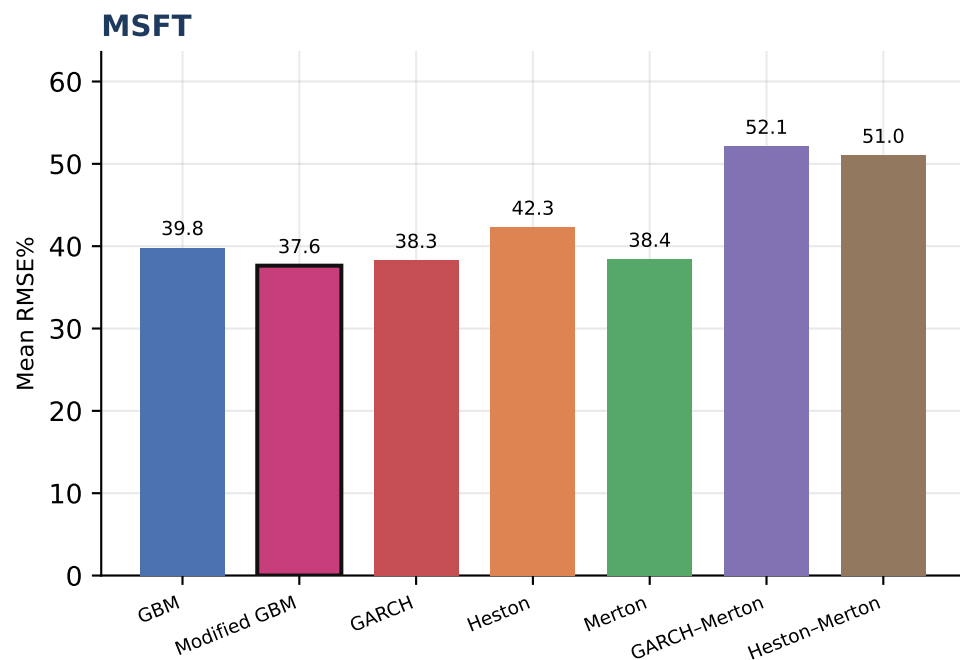
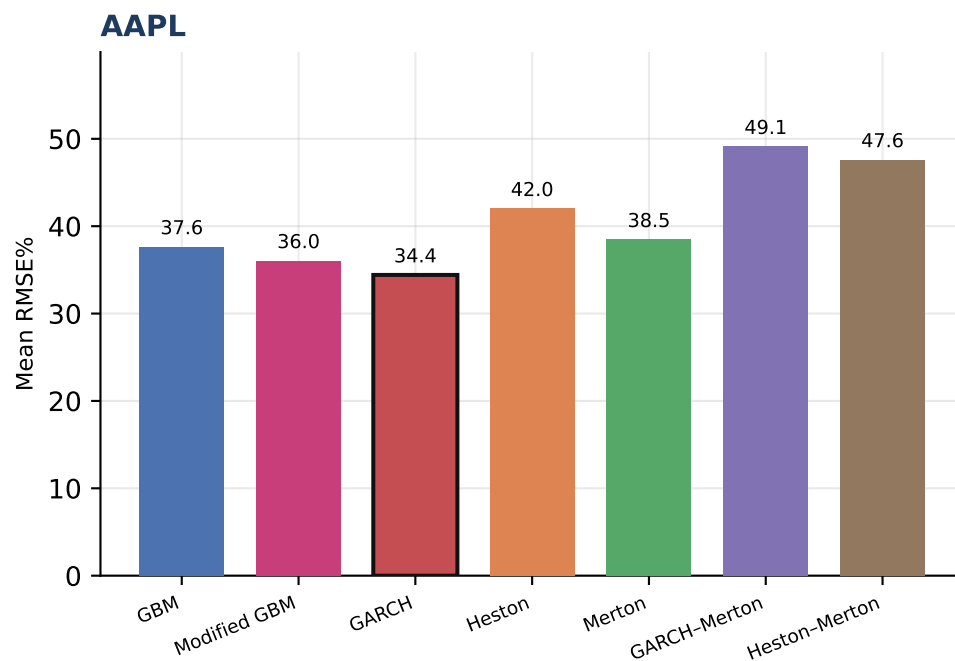
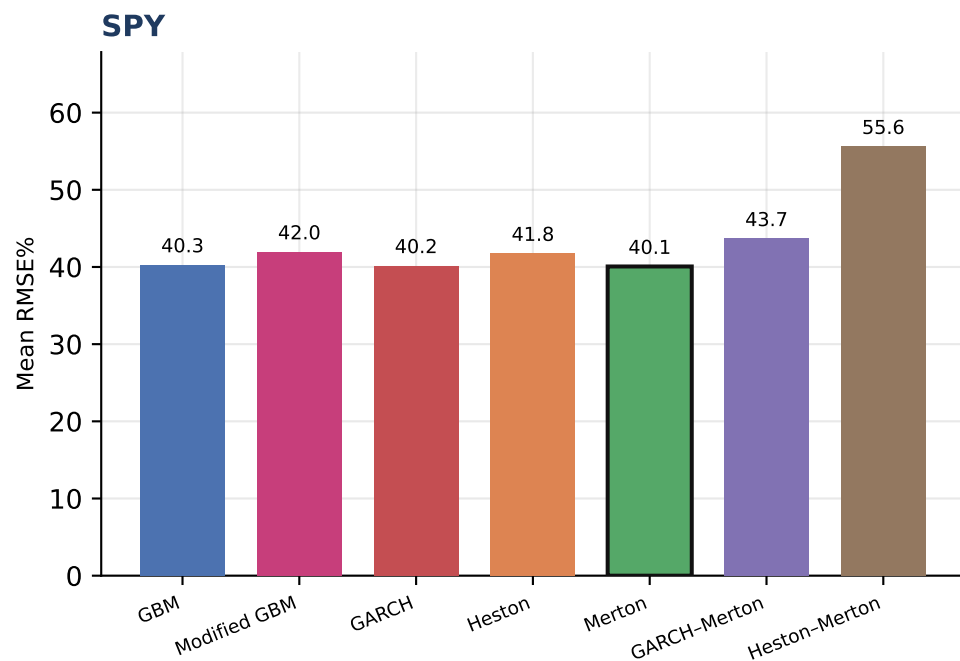
Table S4.4 · AMZN

| Model         | Mean RMSE%   | Median       | # regimes best |
|---------------|--------------|--------------|----------------|
| GBM           | 33.82        | 27.73        | 1              |
| Modified GBM  | 34.21        | 26.79        | 0              |
| <b>GARCH</b>  | <b>28.87</b> | <b>27.69</b> | <b>0</b>       |
| Heston        | 35.16        | 34.77        | 1              |
| Merton        | 33.48        | 26.53        | 1              |
| GARCH-Merton  | 40.50        | 40.09        | 0              |
| Heston-Merton | 46.12        | 48.58        | 1              |

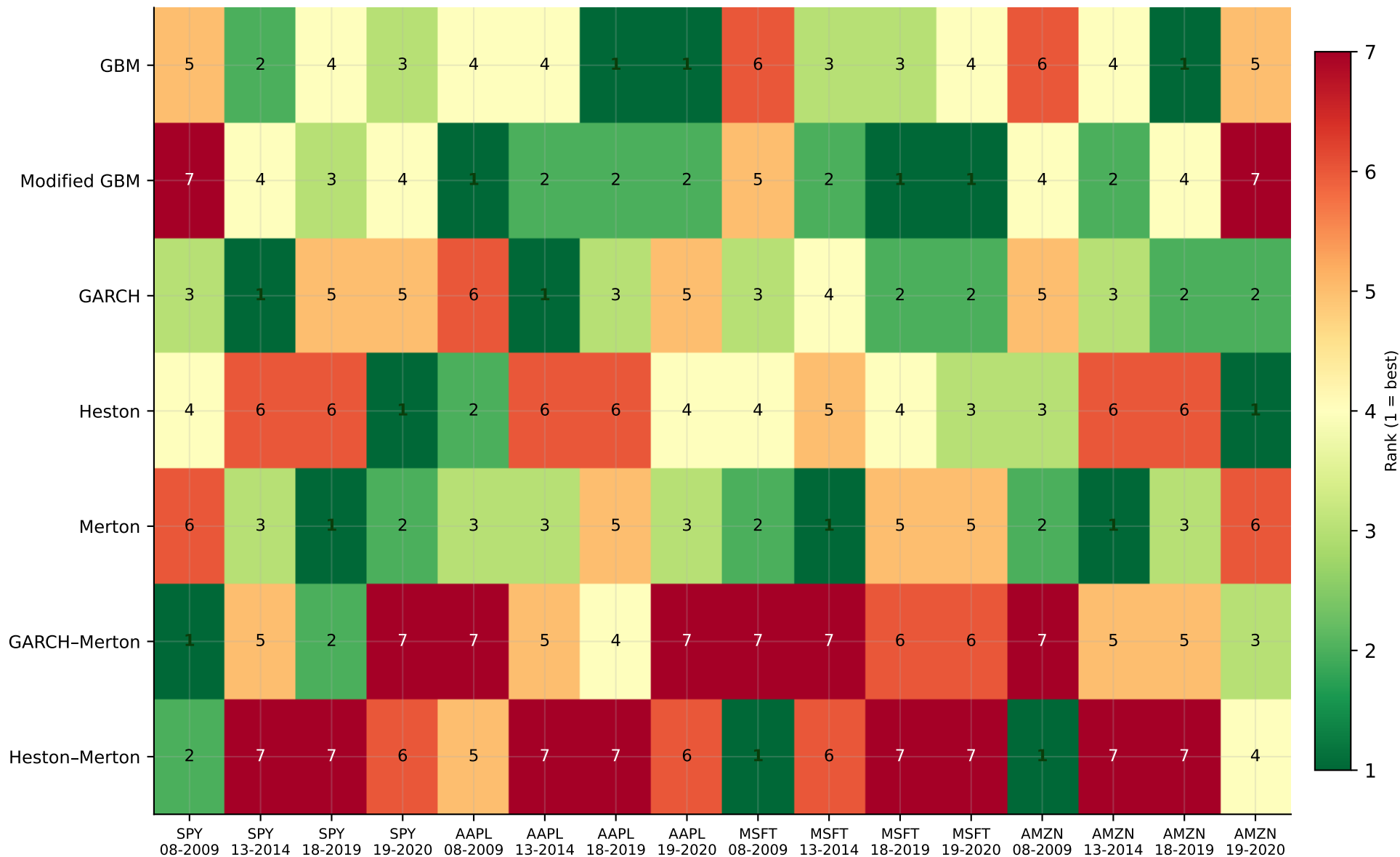
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



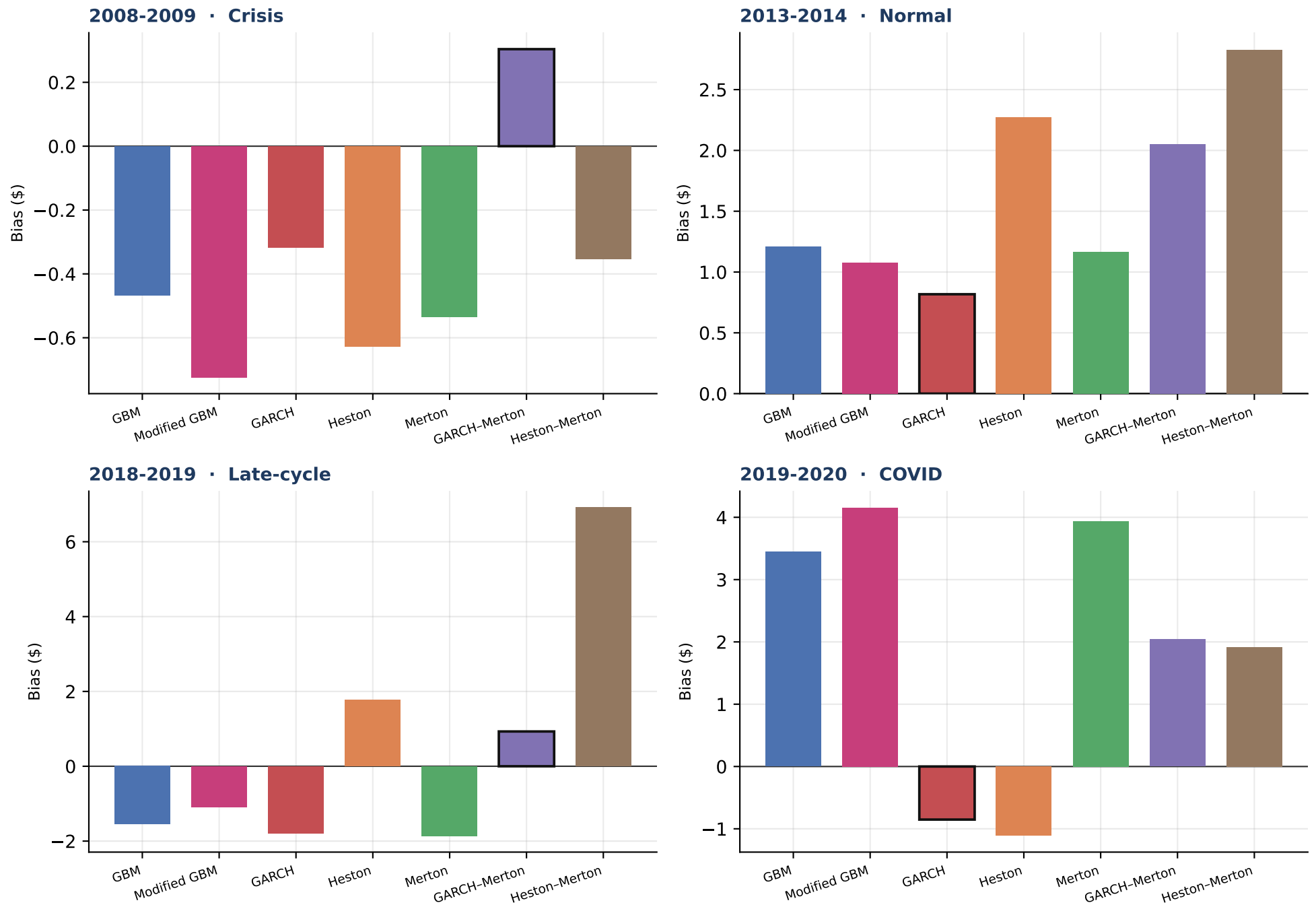
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

### Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, GARCH is the best overall model (mean RMSE% = 35.43; 2 of 16 cells).
- Second overall is Modified GBM (mean RMSE% = 37.46; 3 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: GARCH, GARCH-Merton, GBM, Heston, Heston-Merton, Merton, Modified GBM.
- SPY: lowest mean RMSE% is Merton (40.07); wins 1 of 4 regimes.
- AAPL: lowest mean RMSE% is GARCH (34.42); wins 1 of 4 regimes.
- MSFT: lowest mean RMSE% is Modified GBM (37.63); wins 2 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (28.87); wins 0 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston-Merton (31.42).
- 2013-2014 (Normal): best mean RMSE% is GARCH (41.17).
- 2018-2019 (Late-cycle): best mean RMSE% is Modified GBM (20.89).
- 2019-2020 (COVID): best mean RMSE% is Heston (44.54).
- No-jump models as a group have lower mean RMSE% (37.77) than the jump models in this report (44.68).
- For the overall winner (GARCH), crisis RMSE% is 32.44 vs 41.17 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year fixed calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

### How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all seven models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).

Early-exercise fractions are a model implication, not an accuracy score. American calls on these tenors often have modest early-exercise value.

Notebook: `PhyLib-1.5` · `PhyLib-1.5` is a fixed calibration, 10,000 paths, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. American calls on these tenors often have modest early-exercise value.  
Engine: `V4-Models_result/scripts/run_v4_1p5y_10k_fixed_empirical_study.py` · `payload.json` is the shared number store.